

Minecraft Map Build Guide

Design by Concept_Bytes



Project overview

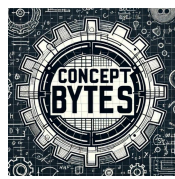
The working minecraft map is a raspberry pi and a display that shows you your real location in google maps with an applied minecraft texture. This project can be 3d printed, but feel free to add your own spin on it. With this build guide I am also providing STL files, and code to run it!

This Build Manual is a work in progress and will be updated continuously as new issues are fixed and features are added. Make sure to always check back for new updates!

Bill of Materials

Item	Quantity	Cost	Link	Notes
Raspberry Pi zero 2W	1	\$21	Link	

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@concept_bytes.



4in Square Display	1	\$59	Link	
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3D Printed materials

All 3D print files are available on my Patreon

Item	Quantity	STL file	Grams used
Base	1	MC_Map_Base.stl	100
Cover	1	MC_Map_Back.stl	32

Code

The code for this project is available at:

https://github.com/Concept-Bytes/MC_Map

Raspberry pi setup

This project was done on a raspberry PI 2 zero W. This model has a 2.4GHz processor which might be critical for running this program. Other models have not been tested. If you try other models let me know the results.

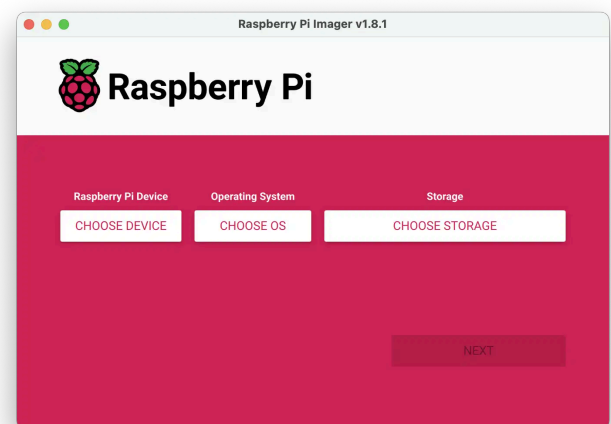
Get Raspberry Pi imager

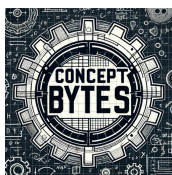
Go to <https://www.raspberrypi.com/software/> to install raspberry pi imager.

Once installed select your device **Raspberry Pi zero 2W**

Select the os which is **Raspberry Pi Os (64 bit)**

Select your storage which should be your SD card





Click **Next**

(This could take some time for your first SD card, after that it should be faster)

Setting up Raspberry pi

Here is a helpful guide to setting up your raspberry pi for the first time:

<https://www.raspberrypi.com/documentation/computers/getting-started.html>

Hardware:

- Insert The SD card into the pi
- Attach a mouse and keyboard
- Set up HDMI to a monitor (if you prefer to SSH into the pi you can do that too)
- Plug into power supply via usb-c

Software

- Connect your Raspberry pi to a wifi network
- Open a terminal by clicking the black icon at the top right
- Run the following commands:

```
sudo apt update
```

```
sudo apt full-upgrade
```

Setting up the waveshare display:

There are a few extra things you have to do to get this screen working that make it a little extra tricky.

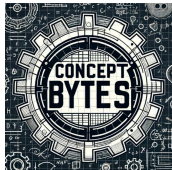
- 1: Modify the config.txt to include the screen config
- 2: Add the display files to /boot/overlays

This guide linked below will walk you through everything you need to know for setting up your display:

[https://www.waveshare.com/wiki/4inch_DPI_LCD_\(C\)](https://www.waveshare.com/wiki/4inch_DPI_LCD_(C))

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Cloning the Map repository

https://github.com/Concept-Bytes/MC_Map

Running the software

```
cd MC_Map/
```

You will need to edit the code to include:

Your coordinates

Google maps API Key which you can get by following:

<https://developers.google.com/maps/documentation/embed/get-api-key>

Once that is done you can run:

```
python map.py
```

And then:

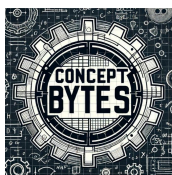
```
python display.py
```

3D printing guide

There are 2 files for this project. I suggest not scaling them as they need to fit the display perfectly.

3d printing settings

file	infil	Layer Height
MC_Map_Base.stl	15%	0.2mm
MC_Map_Back.stl	15%	0.2mm



Assembly

Step 1

Start by acquiring your 4in waveshare display, and Raspberry pi. Buying some heatsinks for the pi is also a great option!



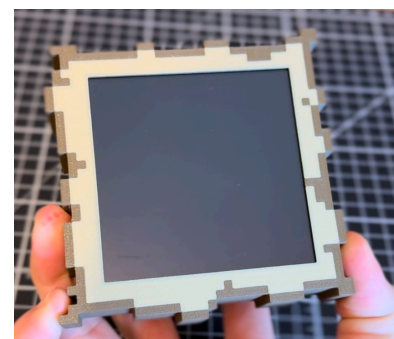
Step 2

You may need to solder on the pin headers to the raspberry pi, do this first. The headers should be on the top of the board, and the board will plug into the back of the display.



Step 3

Your display can not go face down inside the 3D printed base. Thread a micro usb cable through the backing and out of the case, then snap the back on.



Step 4

Now plug in your pi! Remember your Pi needs 5v 2A minimum. The official raspberry pi power cable is the best bet if yours is running slow. Those run at 5v 3A or 15W. This is highly recommended for best performance.